



Figure 1: Ubuntu as a router



Figure 2: Configuration of eth0



Figure 3: Setting up network on eth0

To summarise:

- eth1 = Network adapter connected to the Internet (external)
- eth0 = Network adapter connected to a computer in the same subnet (internal)
- 10.10.6.0 = Subnet for eth0
- 192.168.1.8 = IP address of Host A, any computer in the Internet
- 10.10.6.203 = IP address of eth0.
- 10.10.6.204 = IP address of eth1.
- 10.10.6.205 = IP address of Host B, any computer in the same subnet.

## Configuring network interface cards

Each network interface has to be assigned a static IP address. How to do this differs for the desktop edition and the server edition of Ubuntu. Both the methods are described below. You can refer Figure 2 to 5

**For Ubuntu's desktop edition:** Click on **System Settings** ⇒ **Network** ⇒ **Select Interface** ⇒ **Options**

**For Ubuntu's server edition:** You need to follow the steps given below.

1. Open the terminal, by pressing **Ctrl+Alt+T**
2. Enter the following command to edit the 'interfaces' file:

```
sudo vim /etc/network/interfaces
```

3. Edit the file with the following lines:

```
auto lo
iface lo inet loopback
auto eth0
iface eth0 inet static
address 10.10.6.203
netmask 255.255.255.0
gateway 10.10.6.203
auto eth1
iface eth1 inet static
address 10.10.6.204
netmask 255.255.255.0
gateway 10.10.6.2
```

## Enable IP forwarding

Configure the Ubuntu system so as to initiate routing between two interfaces by enabling IP forwarding:

```
sudo sh -c "echo 1 /proc/sys/net/ipv4/ip forward"
```

Edit `/etc/sysctl.conf`, and add the following lines (for versions up to Ubuntu 10.04):

```
net.ipv4.conf.default.forwarding=1
net.ipv4.conf.all.forwarding=1
```

From Ubuntu 10.10 onwards, it is sufficient to edit `/etc/sysctl.conf` and uncomment:

```
# net.ipv4.ip forward=1
```

...so that it reads as follows:

```
net.ipv4.ip forward=1
```

## IP masquerading

To enable IP masquerading, enter the following set of commands at the terminal:

```
sudo iptables -t nat -A POSTROUTING -o eth1 -j MASQUERADE
sudo iptables -A FORWARD -i eth1 -o eth0 -m state --state RELATED,ESTABLISHED -j ACCEPT
```



Figure 4: Configuration of eth1



Figure 5: Setting up network on eth1